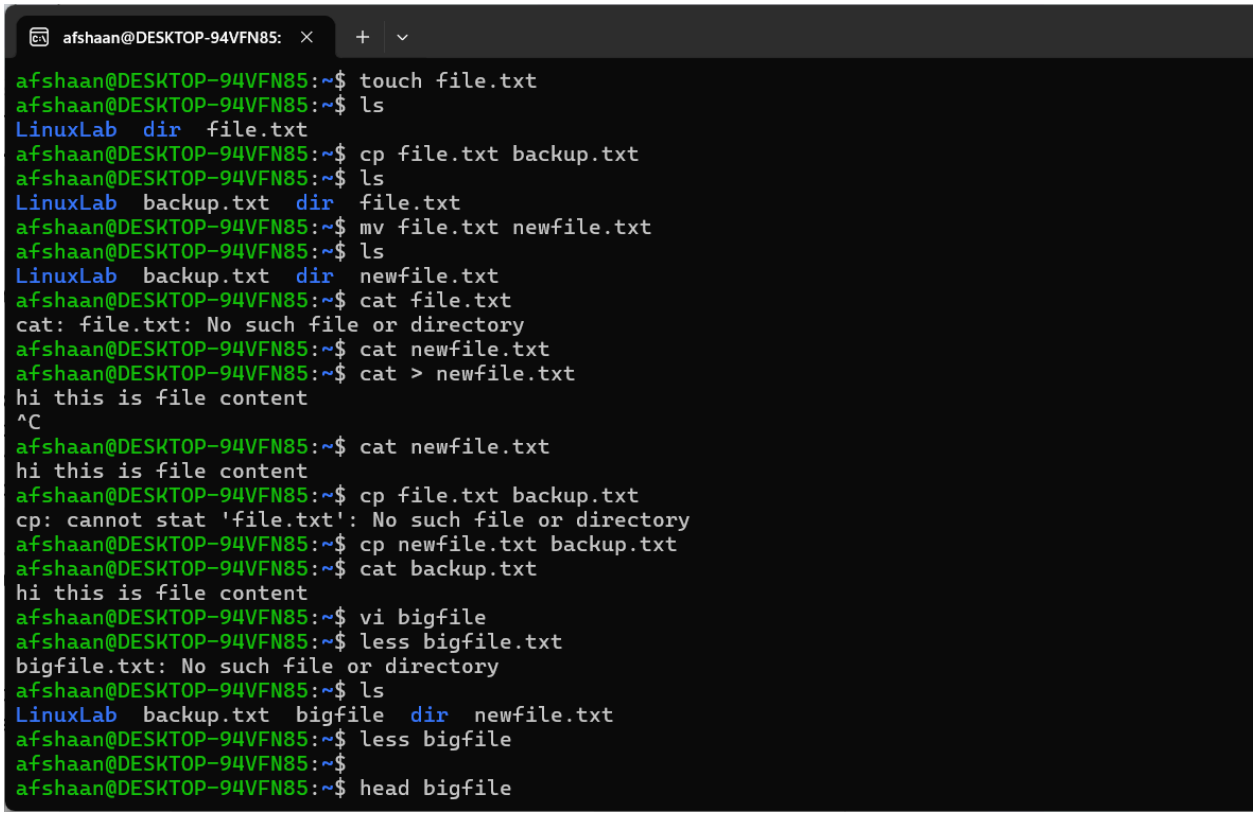
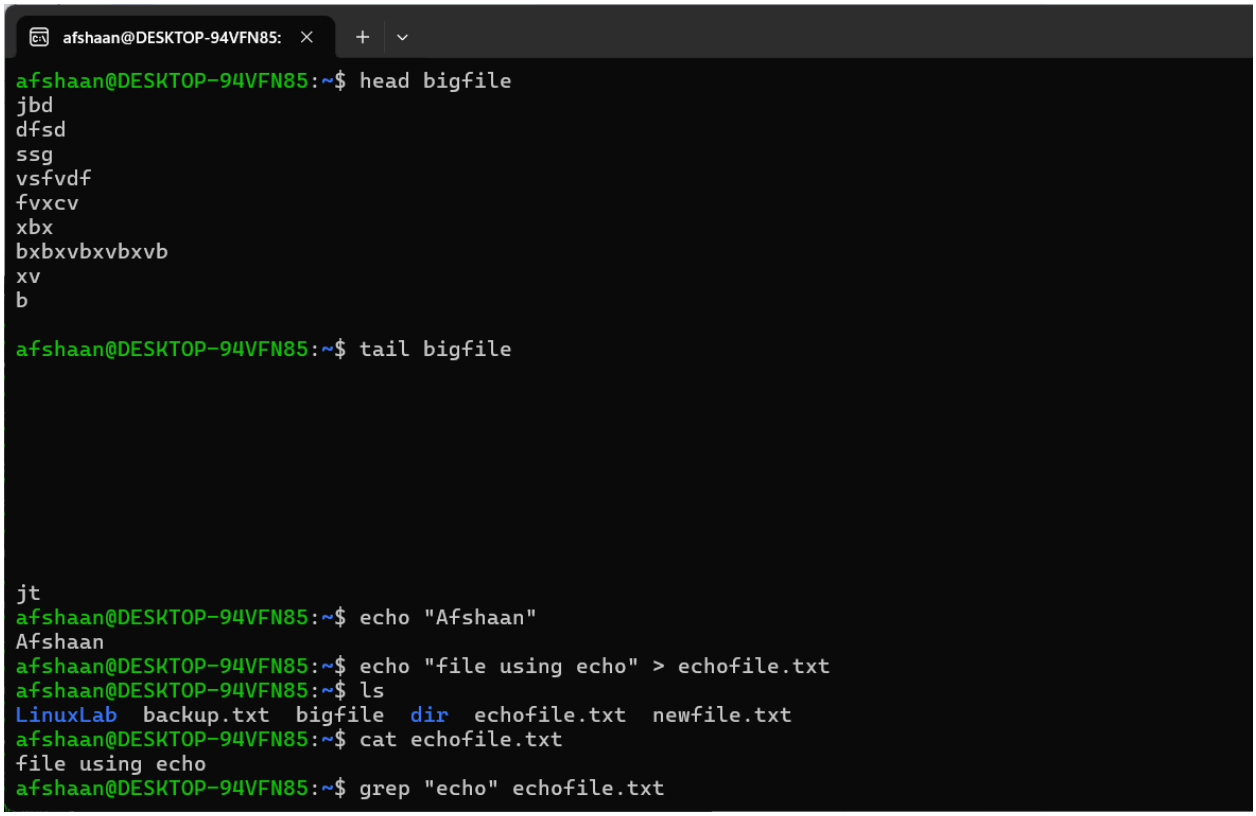
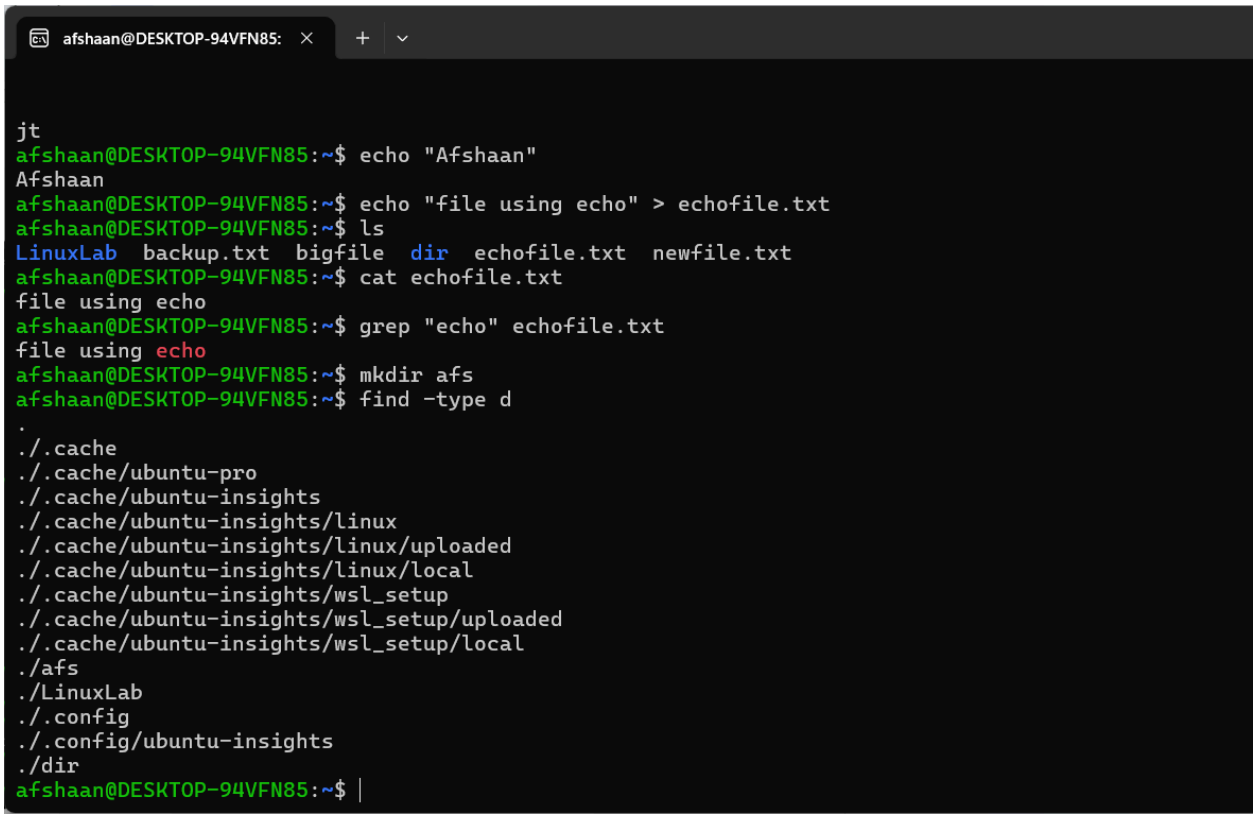

afshaan@DESKTOP-94VFN85:~$ touch file.txt
afshaan@DESKTOP-94VFN85:~$ ls
LinuxLab dir file.txt
afshaan@DESKTOP-94VFN85:~$ cp file.txt backup.txt
afshaan@DESKTOP-94VFN85:~$ ls
LinuxLab backup.txt dir file.txt
afshaan@DESKTOP-94VFN85:~$ mv file.txt newfile.txt
afshaan@DESKTOP-94VFN85:~$ ls
LinuxLab backup.txt dir newfile.txt
afshaan@DESKTOP-94VFN85:~$ cat file.txt
cat: file.txt: No such file or directory
afshaan@DESKTOP-94VFN85:~$ cat newfile.txt
afshaan@DESKTOP-94VFN85:~$ cat > newfile.txt
hi this is file content
^C
afshaan@DESKTOP-94VFN85:~$ cat newfile.txt
hi this is file content
afshaan@DESKTOP-94VFN85:~$ cp file.txt backup.txt
cp: cannot stat 'file.txt': No such file or directory
afshaan@DESKTOP-94VFN85:~$ cp newfile.txt backup.txt
afshaan@DESKTOP-94VFN85:~$ cat backup.txt
hi this is file content
afshaan@DESKTOP-94VFN85:~$ vi bigfile
afshaan@DESKTOP-94VFN85:~$ less bigfile.txt
bigfile.txt: No such file or directory
afshaan@DESKTOP-94VFN85:~$ ls
LinuxLab backup.txt bigfile dir newfile.txt
afshaan@DESKTOP-94VFN85:~$ less bigfile
afshaan@DESKTOP-94VFN85:~$
afshaan@DESKTOP-94VFN85:~$ head bigfile


afshaan@DESKTOP-94VFN85:
afshaan@DESKTOP-94VFN85:~$ head bigfile
jbd
dfsd
ssg
vsfvdf
fvxcv
xbx
bxbxvbxvbxvb
xv
b
afshaan@DESKTOP-94VFN85:~$ tail bigfile
jt
afshaan@DESKTOP-94VFN85:~$ echo "Afshaan"
Afshaan
afshaan@DESKTOP-94VFN85:~$ echo "file using echo" > echofile.txt
afshaan@DESKTOP-94VFN85:~$ ls
LinuxLab backup.txt bigfile dir echofile.txt newfile.txt
afshaan@DESKTOP-94VFN85:~$ cat echofile.txt
file using echo
afshaan@DESKTOP-94VFN85:~$ grep "echo" echofile.txt


afshaan@DESKTOP-94VFN85:
jt
afshaan@DESKTOP-94VFN85:~$ echo "Afshaan"
Afshaan
afshaan@DESKTOP-94VFN85:~$ echo "file using echo" > echofile.txt
afshaan@DESKTOP-94VFN85:~$ ls
LinuxLab backup.txt bigfile dir echofile.txt newfile.txt
afshaan@DESKTOP-94VFN85:~$ cat echofile.txt
file using echo
afshaan@DESKTOP-94VFN85:~$ grep "echo" echofile.txt
file using echo
afshaan@DESKTOP-94VFN85:~$ mkdir afs
afshaan@DESKTOP-94VFN85:~$ find -type d
.
./.cache
./.cache/ubuntu-pro
./.cache/ubuntu-insights
./.cache/ubuntu-insights/linux
./.cache/ubuntu-insights/linux/uploaded
./.cache/ubuntu-insights/linux/local
./.cache/ubuntu-insights/wsl_setup
./.cache/ubuntu-insights/wsl_setup/uploaded
./.cache/ubuntu-insights/wsl_setup/local
./afs
./LinuxLab
./.config
./.config/ubuntu-insights
./dir
afshaan@DESKTOP-94VFN85:~$

1. touch

The touch command is used to create an empty file. If the file already exists, it updates the file's timestamp without deleting its contents.

Examples:

- touch file.txt — Creates an empty file named file.txt.
- touch notes.txt backup.txt — Creates two empty files.
- touch /home/user/report.txt — Creates a file at the specified location.
- touch file1.txt file2.txt file3.txt — Creates three files at once.

2. cp

The cp command is used to copy files and directories from one location to another. The original file remains unchanged.

Examples:

- cp notes.txt backup.txt — Copies notes.txt into a new file named backup.txt.
- cp file.txt /tmp/ — Copies file.txt into the /tmp directory.
- cp file.txt /home/user/Documents/ — Copies a file into the Documents directory.
- cp -r project/ backup/ — Copies a directory and all its contents.
- cp *.txt /tmp/ — Copies all files ending in .txt into /tmp/.
- cp -i notes.txt backup.txt — Asks for confirmation before overwriting an existing destination file.

3. mv

The mv command is used to move files and directories or rename them. Unlike cp, it normally removes the original path after moving the item.

Examples:

- mv old.txt new.txt — Renames old.txt to new.txt.
- mv file.txt /tmp/ — Moves file.txt into /tmp/.
- mv notes.txt /home/user/Documents/ — Moves the file into Documents.
- mv folder1 /tmp/ — Moves a directory into /tmp/.
- mv file1.txt file2.txt /tmp/ — Moves both files into /tmp/.
- mv -i file.txt /tmp/ — Asks for confirmation before overwriting an existing destination file.

4. cat

The cat command is used to display file contents, combine files, or write text into files using redirection.

Examples:

- cat file.txt — Displays the contents of file.txt.
- cat /etc/hosts — Displays the system's hosts file.
- cat file1.txt file2.txt — Displays the contents of both files consecutively.
- cat file1.txt file2.txt > combined.txt — Combines the contents of two files into combined.txt.
- cat > newfile.txt — Allows you to type content into a new file. Press Ctrl+D to finish.
- cat -n file.txt — Displays file contents with line numbers.

Note: Using > overwrites an existing destination file. Use >> to append content instead.

5. less

The less command is used to view large files page by page. It is useful when a file contains too much content to fit on one terminal screen.

Examples:

- less file.txt — Opens a text file for page-by-page viewing.
- less /var/log/syslog — Views the system log file.
- less /etc/passwd — Views user account information.
- less +G file.txt — Opens the file at the end.
- less +F app.log — Follows new content being added to a file, similar to tail -f.

Useful keys while viewing a file:

- Space — Moves forward one page.
- b — Moves backward one page.
- Down Arrow — Moves down one line.
- /error — Searches for the word error.
- n — Moves to the next search match.
- q — Quits the viewer.

6. head

The head command displays the beginning of a file. By default, it shows the first 10 lines.

Examples:

- head file.txt — Displays the first 10 lines.
- head -n 5 file.txt — Displays the first 5 lines.
- head -n 20 app.log — Displays the first 20 lines of a log file.
- head -n 1 names.txt — Displays only the first line.
- head -c 15 file.txt — Displays the first 15 bytes of a file.
- head file1.txt file2.txt — Displays the beginning of both files.

7. tail

The tail command displays the end of a file. By default, it shows the last 10 lines. It is commonly used to monitor log files.

Examples:

- tail file.txt — Displays the last 10 lines.
- tail -n 5 file.txt — Displays the last 5 lines.
- tail -n 20 app.log — Displays the last 20 lines.
- tail -f app.log — Continuously displays new lines added to the file.
- tail -F app.log — Continues following a log file even if it is rotated or recreated.
- tail -c 50 file.txt — Displays the last 50 bytes of a file.

Press Ctrl+C to stop a running tail -f command.

8. echo

The echo command displays text in the terminal. It is also commonly used to write text into files with output redirection.

Examples:

- echo Hello — Prints Hello on the terminal.
- echo "Hello World" — Prints the complete phrase.
- echo "Hello" > hello.txt — Creates hello.txt or overwrites its contents with Hello.
- echo "Welcome" >> hello.txt — Appends Welcome to the end of the file.
- echo "Linux Practice" — Prints a practice message.
- echo $HOME — Displays the current user's home directory.
- echo $USER — Displays the current username.

Important difference:

- > replaces the existing file contents.

- >> adds content to the end of the file.

9. grep

The grep command searches for specific words, phrases, or patterns inside files. It is useful for finding errors, usernames, and particular lines in large files.

Examples:

- grep "error" app.log — Displays lines containing error.
- grep "root" /etc/passwd — Searches for lines containing root.
- grep -i "error" app.log — Searches without distinguishing uppercase and lowercase letters.
- grep -n "error" app.log — Displays matching lines with their line numbers.
- grep -v "error" app.log — Displays lines that do not contain error.
- grep -r "hello" /home/user/ — Searches recursively through files in a directory.
- grep -w "user" file.txt — Searches for the whole word user.
- grep -c "error" app.log — Counts the number of matching lines.

10. find

The find command searches for files and directories based on their names, types, sizes, permissions, and other conditions. You can search within a specific directory or across multiple directories.

Examples:

- find /home -name "notes.txt" — Searches for a file named notes.txt under /home.
- find . -name "*.txt" — Finds all files ending in .txt under the current directory.
- find /home -type f — Finds regular files under /home.
- find /home -type d — Finds directories under /home.
- find /tmp -name "*.log" — Finds files ending in .log under /tmp.
- find . -iname "REPORT.txt" — Searches for REPORT.txt without case sensitivity.
- find /home -type f -size +10M — Finds regular files larger than 10 MB.
- find . -type f -mtime -7 — Finds regular files modified within the last seven days.
- find . -type f -name "*.txt" -delete — Deletes matching text files. Use this carefully and check the search results first.

Note: In find, the dot (.) represents the current directory, while /home specifies the home directory area for all users.